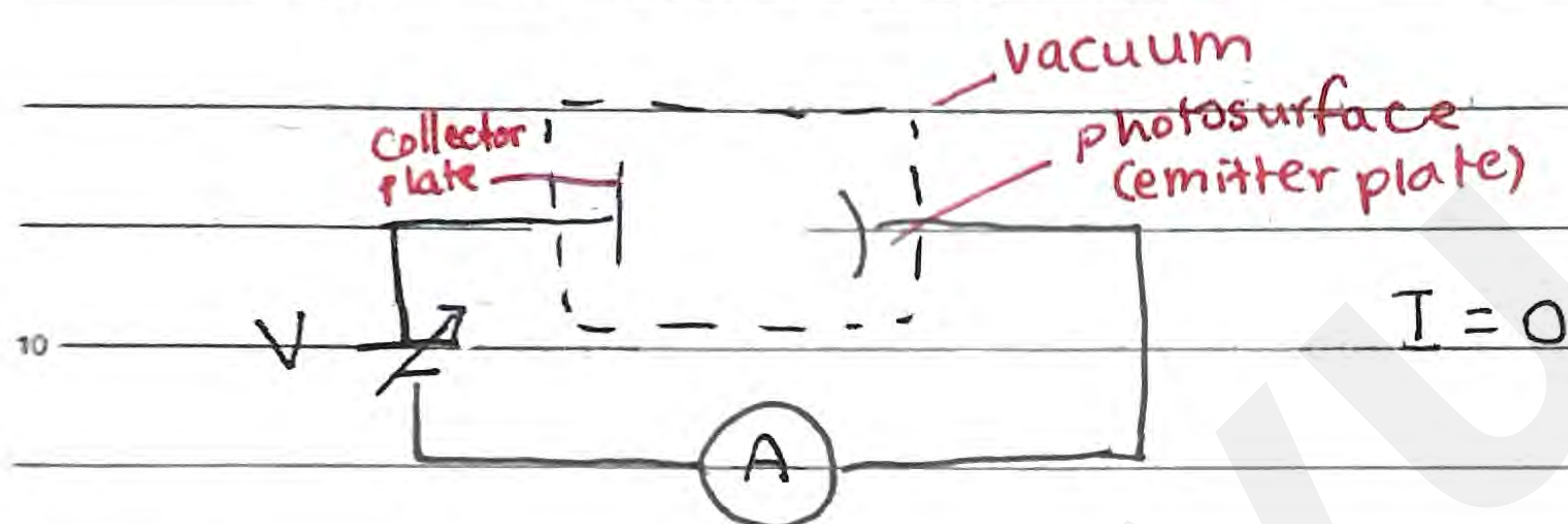


Photoelectric Effect

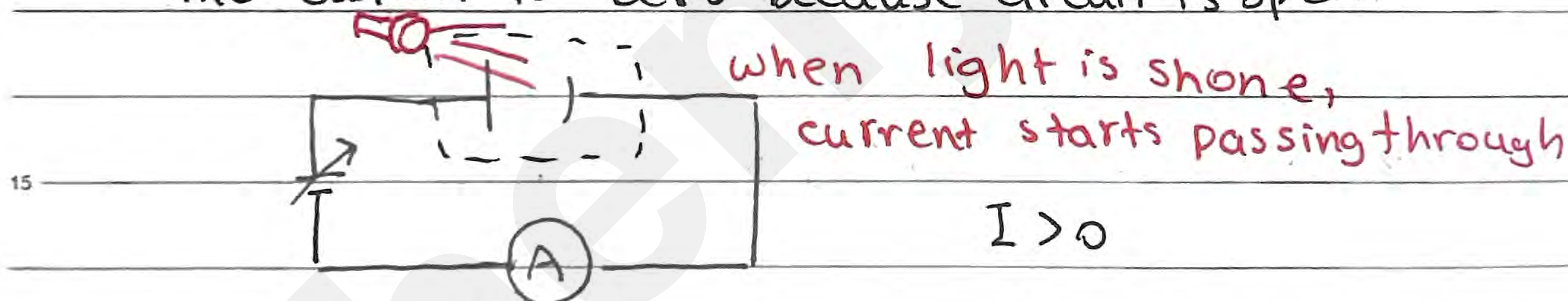
Light, when shone at an open circuit with a certain wavelength, can "complete" a circuit, allowing current to pass through

↳ Discovered in 1800s

↳ The circuit is as follows

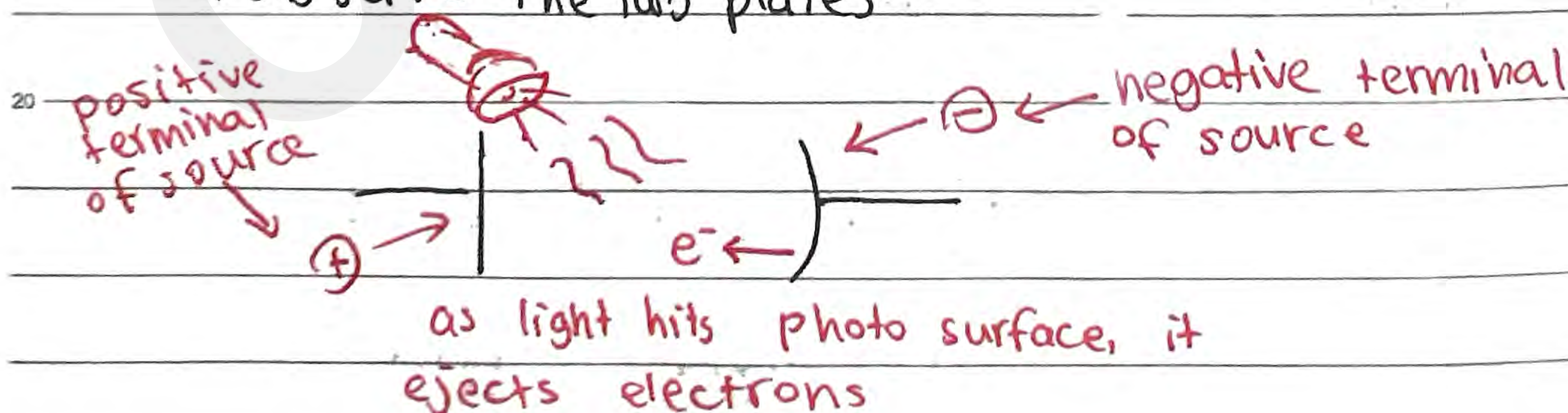


The current is zero because circuit is open



But why? Old theory: Wave theory predictions

↳ Observe the two plates



In a vacuum, electrons face no obstacle,

↳ Paired with charged plates, the field forces electrons to complete circuit

Photoelectric Effect

This explanation is based on classical wave theories $\Rightarrow$ Step-by-step process indicates time delay

$\hookrightarrow$ The scenario happens as long as energy is sufficient to eject electrons

$\hookrightarrow$ Hence, any wavelength works, as long as amplitude and intensity is enough ($E \propto A$)

$\hookrightarrow E_k$ of electrons gets bigger as the amplitude (brightness) increases

But the issue is, this phenomenon only happens for select wavelengths!

Photon Model Predictions (Einstein)

$\hookrightarrow$ 1905, based on light as photon

$\hookrightarrow$ Photon - Smallest bundle or quanta of Electromagnetic radiation

$\hookrightarrow$ A cutoff frequency of light is required in order to eject the electrons from the emitter plate

$\hookrightarrow$ New theory means frequency of light matters

$\hookrightarrow$ Brighter light means more electrons are ejected but does not affect E_k

$\hookrightarrow E_k$ is independent of amplitude. Instead, $A \propto \# \text{ of } e^-$

$\hookrightarrow$ Higher frequency of light produces more energetic electrons

$\hookrightarrow f \propto E_k$

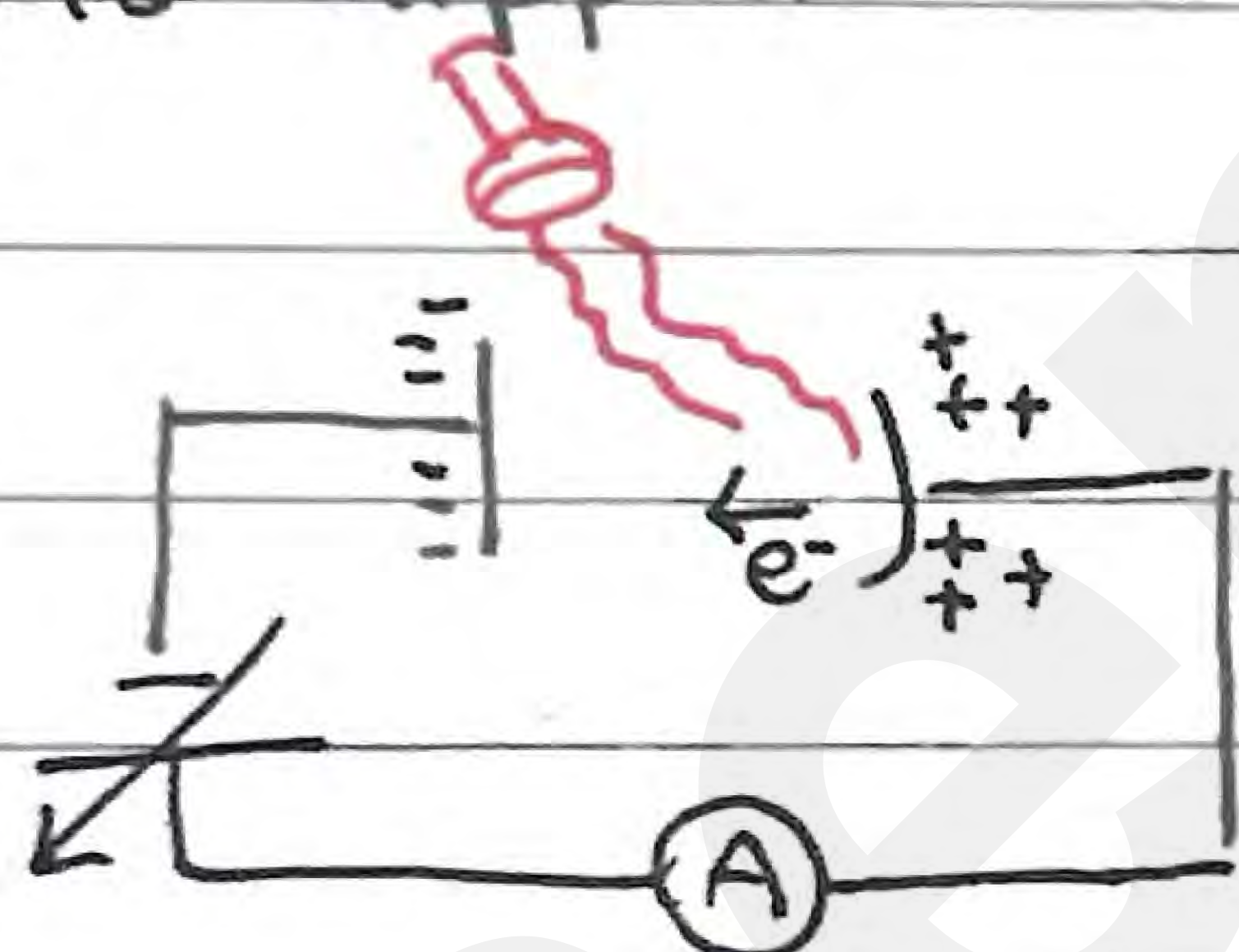
Photon Model Predictions

Additionally, there is no time delay in the phenomenon

↳ After light hits photosurface, current begins instantly

↳ Energy transferred instantly, contrasting old wave theories

Let's consider the case where the voltage source polarity is flipped



The photons transfer energy to the photosurface material

↳ It first overcomes energy to eject electrons ($> \text{Ionization } E$)

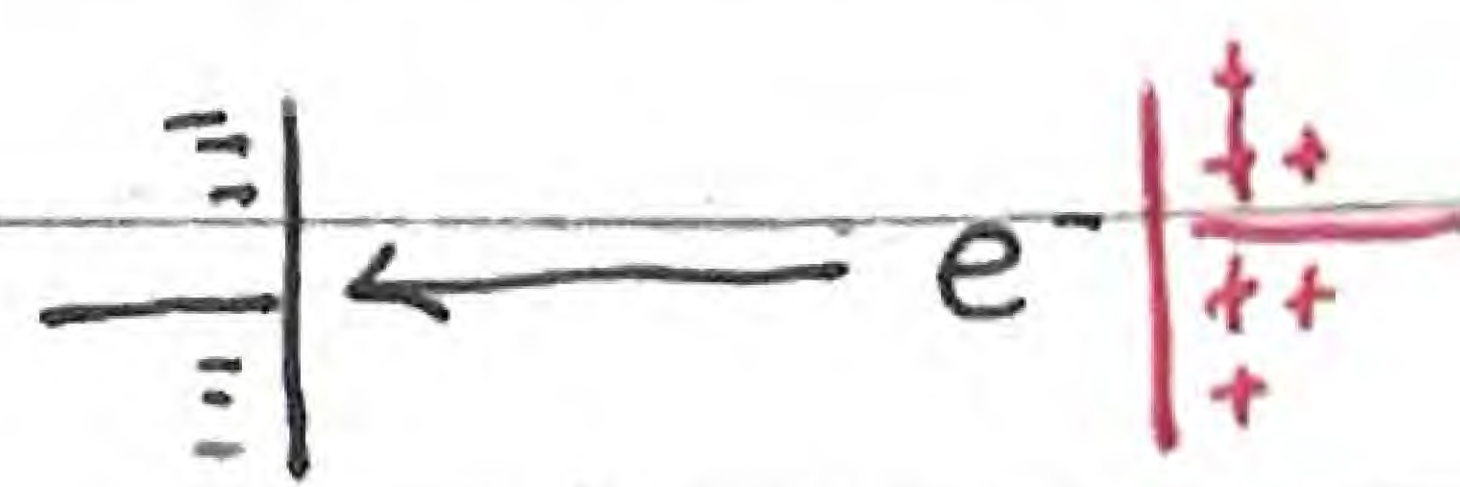
e^- are ejected in all directions (so probabilistically, the wanted direction must also exist)

↳ The excess energy is reflected as kinetic energy

But, since polarity is flipped, e^- 's are repulsed from collector plate, so kinetic energy must also be enough for current to flow



insufficient



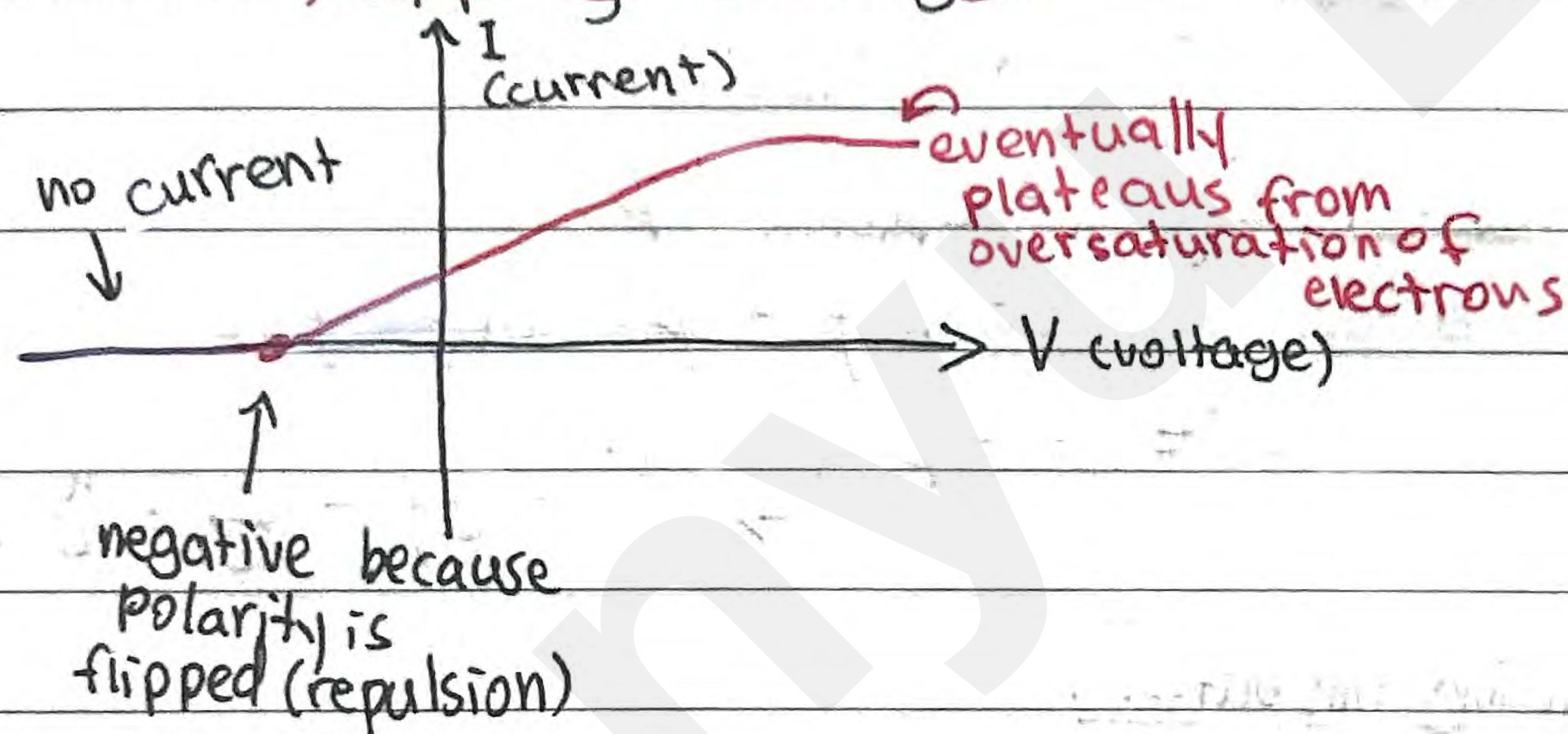
enough

Photoelectric effect

But we have a variable voltage source

↳ That potential difference controls the repulsion the electron experiences

↳ The minimum voltage at which there exists current (electrons barely complete circuit) is the Stopping Voltage



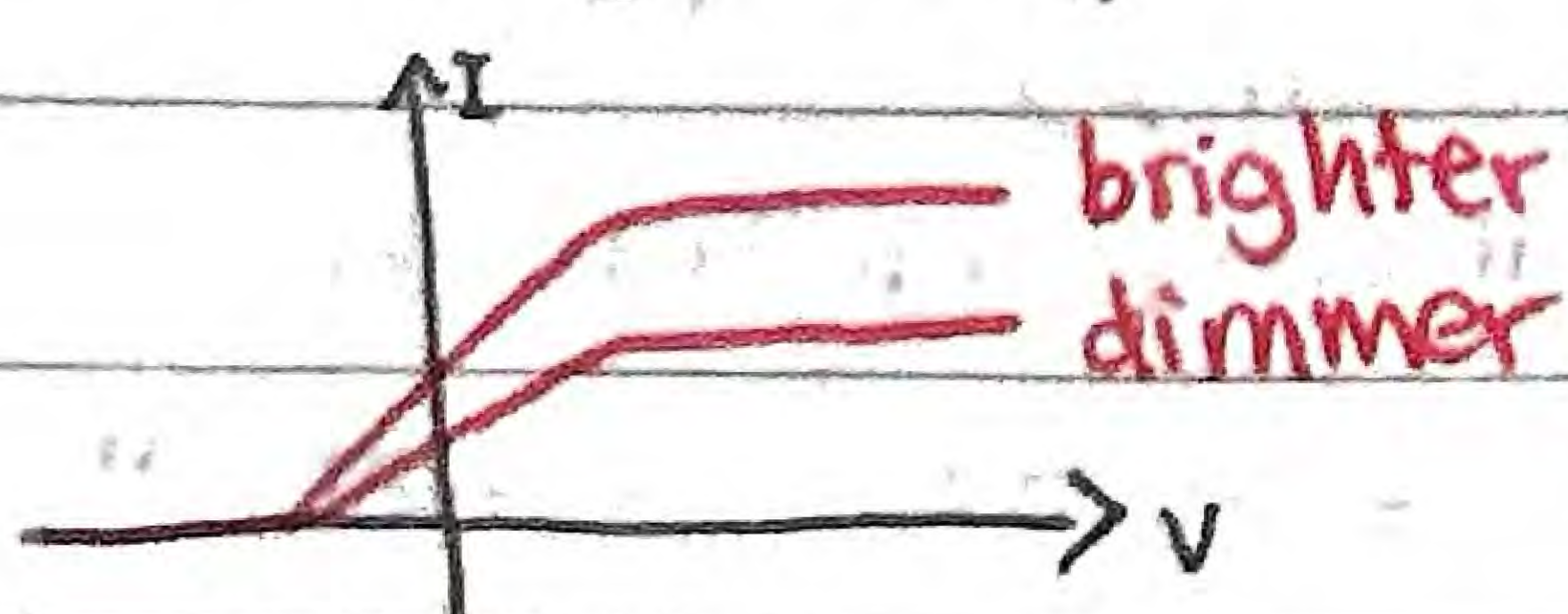
Stopping voltage is the minimum potential difference

↳ Hence maximum kinetic energy is given by

$$E_k = qV \Rightarrow E_{k, \max} = e \cdot V_s$$

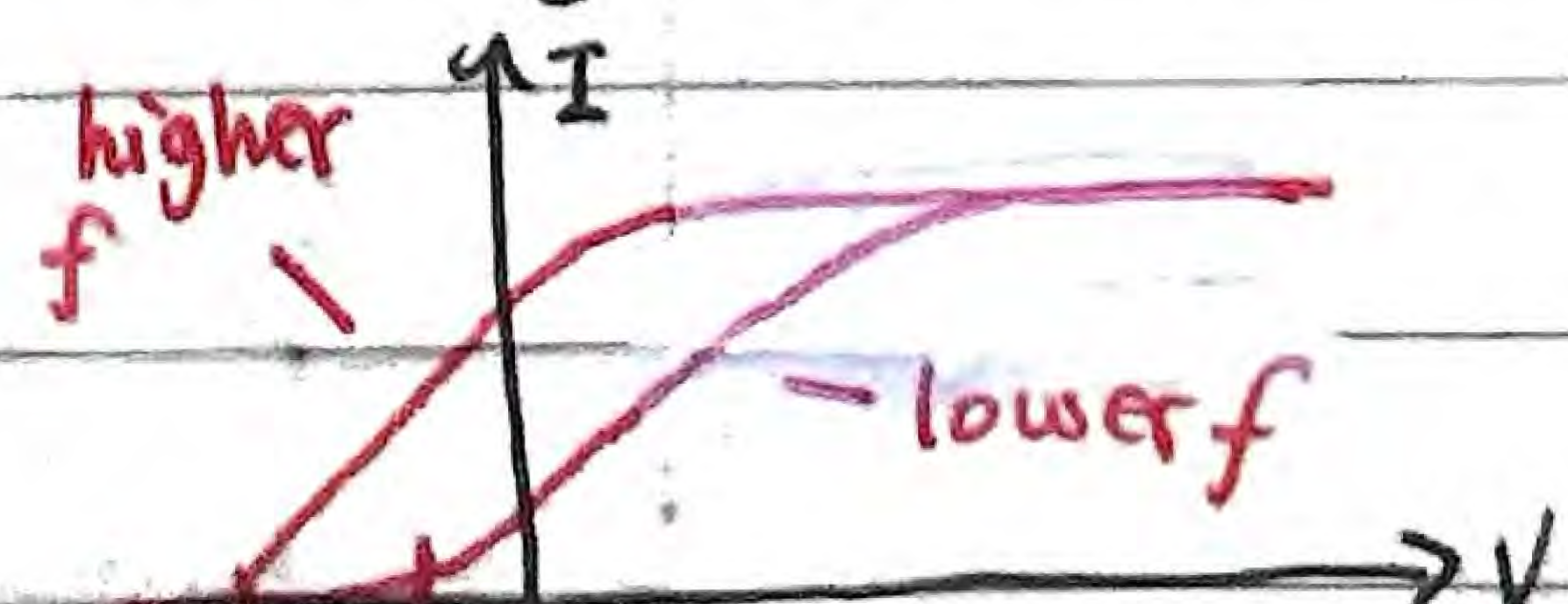
We've changed properties of the voltage source, but how about properties of light and photons?

Current vs Voltage
(varying intensity)



- Same V_s
- Greater slope/max. for brighter (more e^- ejected)

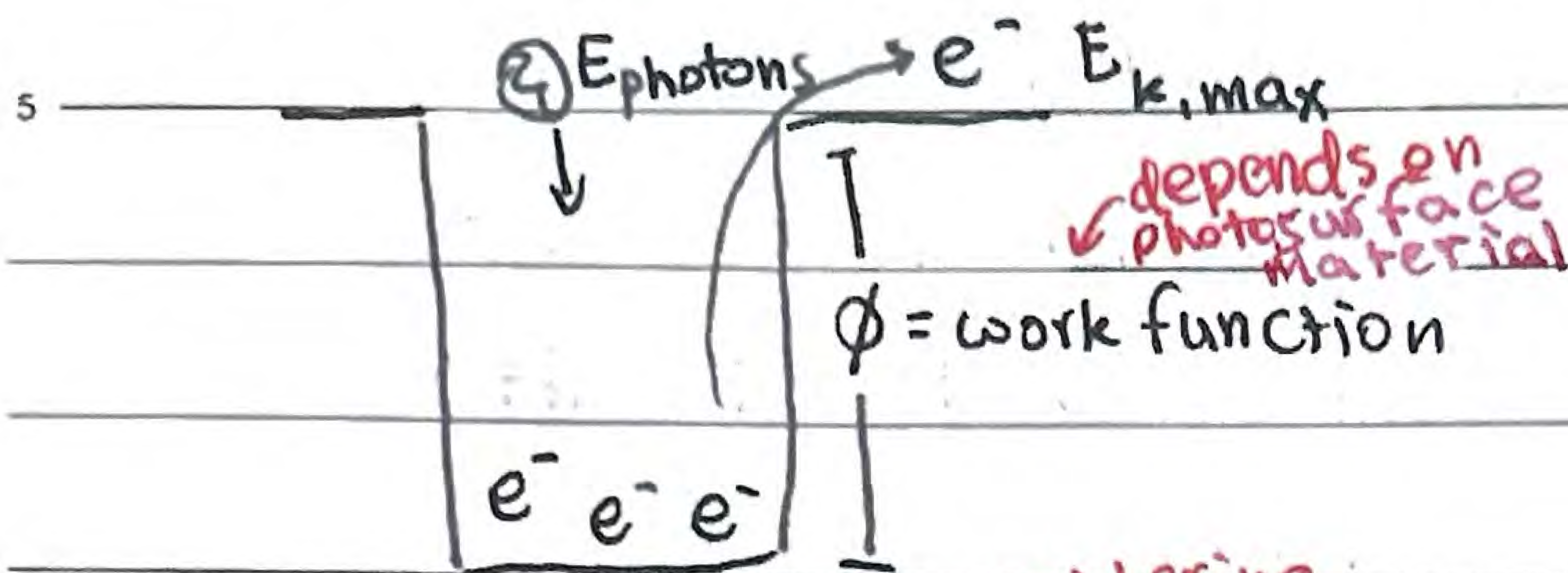
Current vs Voltage
(varying frequency)



- Different V_s (E_k different)
- But same max/slope (same # of e^- ejected)

Photoelectric Effect

observing the photosurface more specifically...



Work function

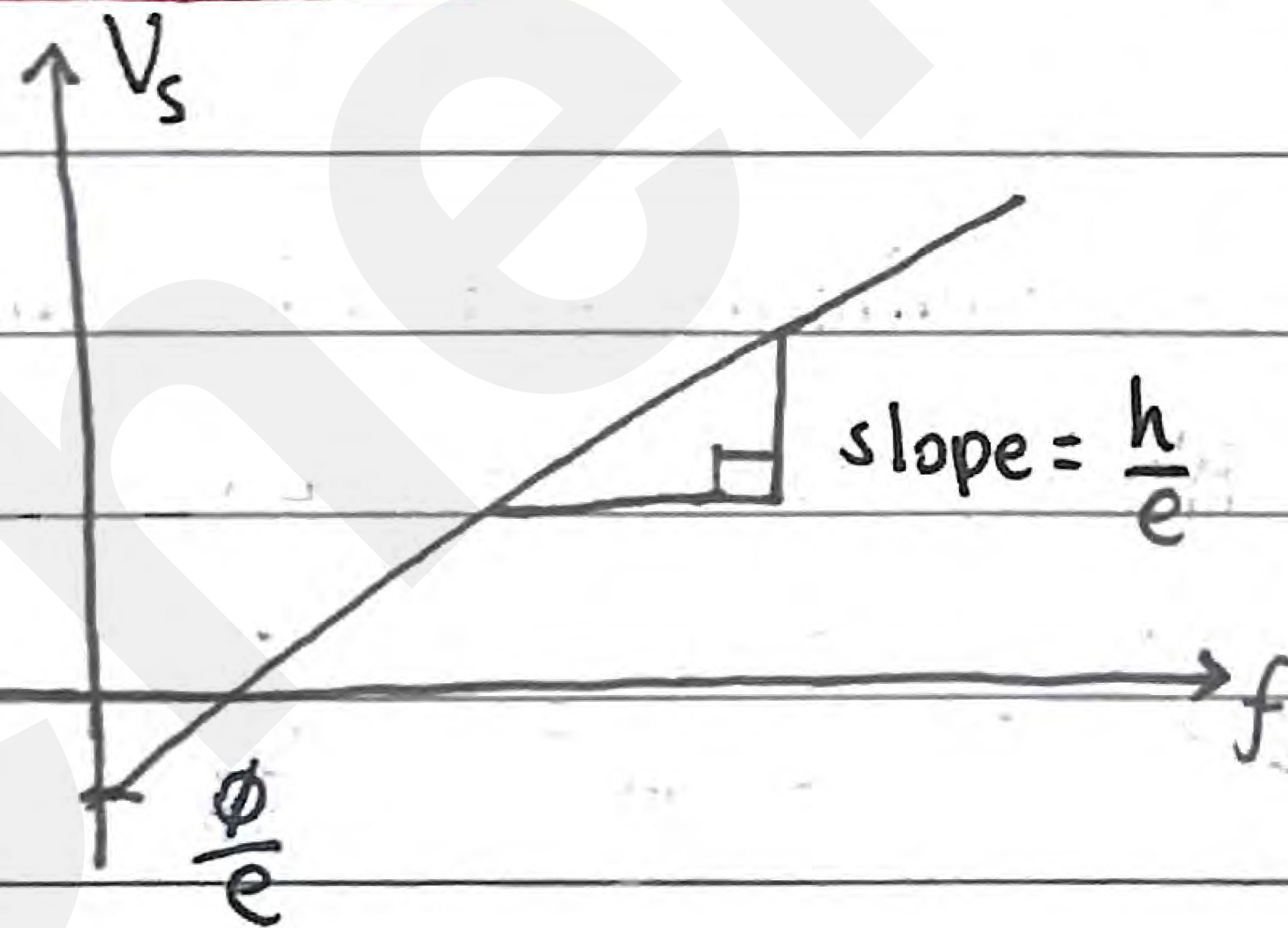
needs to be overcome
before e^- are free

only considering
photosurface,
no energy gained
after

Hence... $E_{k, \text{max}} = E_{\text{photon}} - \phi$
 $= h \cdot f - \phi = h(f - f_0)$

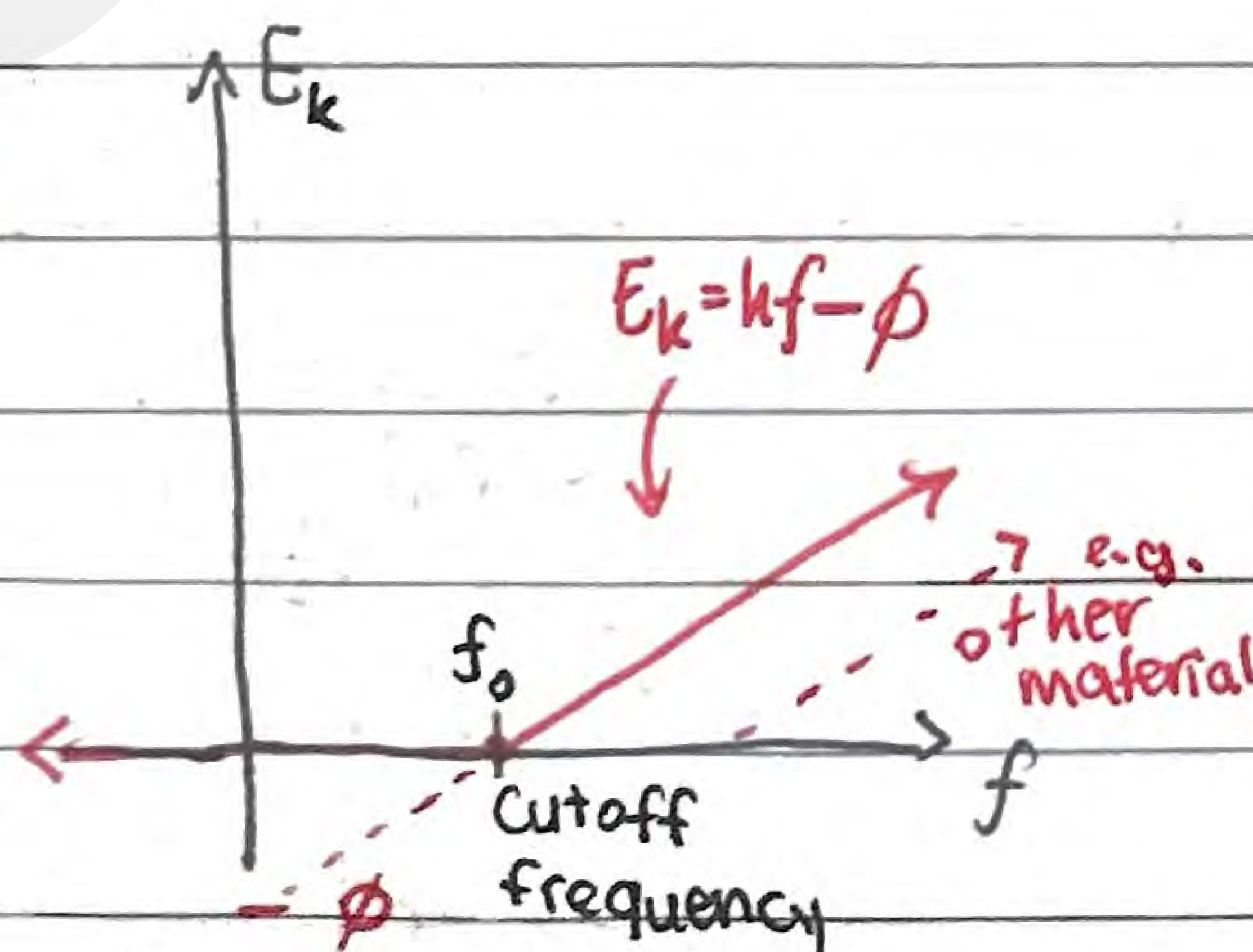
$$eV_s = hf - \phi \rightarrow V_s = \frac{h}{e}f - \frac{\phi}{e} = \frac{h}{e}(f - f_0)$$

Graphing this out...



Sometimes power can be given
 $\hookrightarrow$ You get energy per second
 $\hookrightarrow$ And photons per second

and...



Different materials have
different cutoff frequencies
(different ϕ)
 $\hookrightarrow$ But every material has same
slope (planck's constant h)

this is where
the constant is obtained

Photoelectric Effect

E.g. A copper plate is illuminated by a beam of monochromatic light of wavelength 580 nm and power 0.5 mW. The work function of copper is 4.7 eV

a) Calculate the # of photons incident on the copper plate in one second

$$E_{\text{photon}} \approx 3.4 \times 10^{-19} \text{ J} \Rightarrow P_{\text{total}} = 0.5 \times 10^{-3} \text{ J s}^{-1}$$

$$\text{So \# photons/s} = P_{\text{total}} \div E_{\text{photon}} \approx 1.5 \times 10^{15} \text{ photons s}^{-1}$$

b) Show that no electrons will be emitted

$$E_{\text{photon}} \approx 2.1 \text{ eV} < \text{work function } \phi$$